Affine parameters and a linear CRT selector

Explicit quartic equations for syntactic consistency of ZFC

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Developed with AI assistance; ordinary proofs and exact finite certificates

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Abstract

We simplify the direct numerical proof-calculus construction in the pinned ProveIt repository. The main polynomial has 238 unrestricted integer variables, degree four, 174 quadratic residuals, 5 479 collected monomials, and coefficient height 7 968. A companion uses 237 integer variables. All coefficients and residuals are supplied literally, and both principal residual lists are printed here. The ordinary mathematical proof, composed with the retained C222/ZF/ZFC calibration, establishes in ZFC that each equation has no integer solution if and only if Con(ZFC).

The new mechanism recovers order from already present signed power certificates. It permits affine parameters for the radix, the beta multiplier, and the proof horizon. An elementary least-common-multiple bound also replaces the squared binomial selector offset by its unsquared growth register. These changes remove witnesses without introducing a new power request or a new logical encoding. This is an improvement within the direct small-coefficient quartic family, not a global minimum, a consistency decision, or a new end-to-end proof-assistant formalization.

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1 The result and its precise meaning

The requested equivalence concerns *syntactic* consistency. Fix the repository's recursively presented ZFC axiom predicate and its finite proof relation. Then Con(ZFC) means that no finite derivation of falsity from those axioms exists. We construct ordinary polynomials with fixed integer coefficients, not polynomials whose coefficients depend on an undecided consistency question.

For the main profile let $R_0, \dots, R_{173}$ be the complete quadratic residuals in Appendix A. Define

$$P_{238}(x_0, \dots, x_{237}) = \sum_{j=0}^{173} R_j(x_0, \dots, x_{237})^2. \tag{1}$$

Every coordinate ranges over $\mathbb{Z}$. The finite sum introduces no additional unknown. The endpoint named `primary` in `ENDPOINTS.json` identifies the canonical literal residual list; `expanded_polynomial.json` supplies every collected coefficient of the same polynomial.

Theorem 1.1 (Main equivalence)**.** *For the polynomial in* (1)*, and for every companion in the endpoint table, the following equivalence has an ordinary proof formalizable in ZFC:*

$$\mathrm{ZFC} \vdash [\neg\exists \boldsymbol{x} \in \mathbb{Z}^n \; P_n(\boldsymbol{x}) = 0] \iff \mathrm{Con(ZFC)}.$$

No assumption of consistency or inconsistency is used to prove the equivalence.

1.1 Comparison and the chosen notion of simplicity

Construction	Variables	Rows	Monomials	Height
Inherited factorial-free	242	178	5 018	7 968
Inherited factorial-free, dense	241	177	12 774	7 968
Incoming shared factorial error	240	177	21 482	6 144
Incoming normalized residues	244	176	5 212	3 968
Affine radix and beta	239	175	5 017	7 968
Affine horizon, balanced (main)	238	174	5 479	7 968
Affine horizon, fewer variables	237	173	14 129	7 968

Height is the largest absolute coefficient *after* collecting the whole quartic. Residual height is a different quantity. The number of residuals is not an arithmetic circuit operation count. All comparisons in the table use unrestricted integer variables and degree four.

The comparison snapshot is ProveIt commit `7a8da414376a80b90ecf799f37ceb26fb9ac674e`. Its seven incoming ZIPs were inspected in addition to the integrated collection. In particular, the incoming Shared Factorial Error construction already had a 240-integer quartic, and the Factorial-Free Binomial CRT construction already removed factorial authenticity. We credit and retain those results rather than presenting them as contributions of this report.

The direct family is attractive because its coefficients are inspectable small integers and its logical source is a numerical proof calculus. Other repository constructions already have far fewer variables using different coefficient architectures. The general literature also has universal reductions with eleven integer unknowns [6]. Consequently no unrestricted variable-count record or metric-independent "simplest equation" is claimed here.

2 The retained compiler boundary

The predecessor compiles a finite C222 refutation into bounded arithmetic, then uses a binomial CRT certificate and shared Pell certificates to obtain a fixed quadratic system. We retain its local numerical rule polynomial, all eighteen power requests, all fourteen principal-pair orientation conditions (with an invertible error-basis change), the shared remainder norm, and the exact integral projection machinery.

The logical calibration is the composition of the existing interfaces

`KEndpoint.c222_iff_zf_refutation`,

`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first identifies derivability of the falsity code in C222 with a finite ZF refutation; the second identifies finite ZF and ZFC refutations. The direct construction uses this calculus itself. It does not encode a K215 run and does not specialize Jones's universal polynomial.

For clarity, variable exponentiation and binomial notation in the proof below describe the *semantics* of ordinary polynomial certificates. Neither operation occurs as a primitive in the delivered quartics. All requested powers are implemented by the retained Pell graphs.

The principal semantic registers are:

Register	Meaning after its certificate is authenticated
t	padded number of local proof rows
F	large evaluation point for label multiplicities
y	beta modulus scale
x	common beta code and amplification exponent
B	binomial modulus selecting every proof row
N	B^x
$C, A_1, \dots, A_8$	actual CRT representatives
H	$2 + C + \sum_{i=1}^{8} A_i$
q	radix $(BH)^x$

Some stored coordinates are translated versions of these registers. Semantic names alone do not specify that translation; the exact restoration certificates do.

3 An unsquared binomial selector

Let $L_t = \operatorname{lcm}(1, \dots, t)$. The factorial-free predecessor proves its CRT selector theorem whenever the offset R satisfies $R \geq L_t^2$. It used $R = F^2$. Its growth register is already large enough to take $R = F$ instead.

Lemma 3.1 (Elementary lcm estimate)**.** *For every $n \geq 1$, $L_n \leq 4^{n-1}$.*

Proof. Put $m = \lceil n/2 \rceil$. For any prime p, at most one power of p lies in $(m, n]$. Thus L_n / L_m is squarefree. If p divides this quotient, its largest power $p^a \leq n$ lies in $(m, n]$, and the corresponding Legendre summand in $v_p\binom{n}{m}$ is one. All other summands are nonnegative. Therefore $L_n / L_m \mid \binom{n}{m}$. For $n \geq 2$, each binomial coefficient is at most the sum of coefficients of the same parity, namely 2^{n-1}. Starting from $L_1 = 1$, induction gives

$$L_n \leq 4^{m-1} 2^{n-1} \leq 4^{n-1}.$$

For even n the first exponent of two is $2n - 3$; for odd n it is $2n - 2$. Both give the stated bound. □

This elementary exponential estimate belongs to the classical lcm and binomial-coefficient toolkit [4, 5]. Our contribution here is its use to delete the squared selector register.

Corollary 3.2. *Each of the following choices satisfies $F > L_t^2$:*

$$F = [2(t+1)]^t, \quad t \geq 8,$$
$$F = (t+2)^t, \qquad t \geq 16.$$

Proof. Lemma 3.1 gives $L_t^2 \leq 16^{t-1}$. In the first case the base is at least 18, and in the second it is again at least 18. □

We recall exactly the selector property being used. If $B = \binom{R+t}{t}$ and $R \geq L_t^2$, then every row $i \in \{1, \ldots, t\}$ has a full prime power $p^e \| B$ such that $p^e \mid R + i$ and $p^e > t$. Conversely, every full prime power of B divides at least one $R + i$. Both directions are necessary: the first ensures soundness for every row; the second makes the reverse CRT assignment possible at every prime of B.

For completeness, the first assertion follows by choosing $v = v_p(R + i) > 2a$, where $a = \lfloor \log_p t \rfloor$. Such a prime exists because $R + i > L_t^2$. For $j \neq i$, $v_p(R + j) = v_p(j - i)$, so

$$e = v_p(B) = v - d, \qquad d = v_p\left(t\binom{t-1}{i-1}\right) \leq a.$$

Hence $e > a$ and $e \leq v$. The displayed bound on d follows directly from Legendre's formula: its summands are zero or one and vanish above $p^a \leq t$. For the reverse assertion, let $v = \max_i v_p(R + i)$. The valuation of B is a sum of zero-one carry terms, all zero above level v; hence $e \leq v$, and a maximizing row works.

The new first digit is therefore

$$B = \binom{F+t}{t}, \qquad B \mid C + F + 1. \tag{2}$$

Both occurrences of the offset change together. The corresponding raw quadratic row $R - F^2 = 0$ becomes affine, $R - F = 0$, and its coordinate is eliminated exactly.

4 Signed powers can recover order

4.1 The retained power interface

Write $X_k(A), Y_k(A)$ for the Pell polynomials, with $X_0 = 1, Y_0 = 0$ and

$$X_{k+1} = AX_k + (A^2 - 1)Y_k, \qquad Y_{k+1} = X_k + AY_k.$$

For signed indices use $X_{-k} = X_k$ and $Y_{-k} = -Y_k$. After the principal pairs have been oriented, a request for $m = n^k$ has modulus $T_n = 2An - n^2 - 1$. The shared square frame bounds every actual base, index, and output atom in absolute value by W, and gives $|A| > W^W$.

Lemma 4.1 (Nonunit bases force nonnegative exponents)**.** *Under the retained oriented Pell interface, if $|n| \geq 2$, a requested integer output forces $k \geq 0$ and $m = n^k$, even when n, A, m, k were initially signed.*

Proof. The Pell recurrence gives $m \equiv n^k \pmod{T_n}$ for $k \geq 0$. The absolute difference is smaller than $|T_n| > W^W + W$, so it is zero. If $k = -j < 0$, the inverse-unit recurrence gives $mn^j \equiv 1 \pmod{T_n}$. The inverse is legitimate because $T_n \equiv -1 \pmod{n}$. The absolute error is less than $W^W + 1$, so $mn^j = 1$ in $\mathbb{Z}$, impossible when $|n| \geq 2$ and $j \geq 1$. □

The zero and unit bases are genuine exceptions. The following progressions are chosen specifically to avoid applying the lemma to them.

4.2 Affine radix and beta multiplier

Replace the predecessor's two square parameters by

$$q = 4z + 1, \qquad y = F(2v + 1), \qquad z, v \in \mathbb{Z}. \tag{3}$$

The expression y is now bilinear. The proof cannot initially assume that either q or y is positive. The key unconditional fact is $|q + 1| = |4z + 2| \geq 2$.

Retain the two beta endpoints and the self-radix request:

$$x = F + 1 + (y + 1)u_0, \tag{4}$$
$$x = 1 + [1 + (t + 1)y]u_t, \tag{5}$$
$$q = (BH)^x. \tag{6}$$

Also retain the existing power request with base q and exponent $y + 1$. No extra request is needed for this argument.

Proposition 4.2 (Acyclic sign recovery)**.** *Suppose the signed power interface is available, $t \geq 8$, $F \geq 10^8$, and the unchanged raw digit and endpoint equations hold with (3). Then the system forces*

$$C, A_i \geq 0, \quad H \geq 2, \quad B \neq 0, \quad x > 0, \quad |q| \geq 3, \quad y \geq F,$$

and

$$x \geq F + 1, \qquad x \geq 2 + (t + 1)y, \qquad |q| = |BH|^x \geq 2^x, \qquad |B| < |q|/2. \tag{7}$$

Proof. The numerator requests at base $q + 1$ and Lemma 4.1 first give nonnegative actual CRT representatives. Hence $H \geq 2$. The raw index row $Bh_B = C + F + 1$ excludes $B = 0$, so $|BH| \geq 2$. Its request gives $x \geq 0$ and the exact equality (6).

Before knowing the sign of y, its odd multiplier gives $|y| \geq F$. Thus $|1 + (t + 1)y| > 1$, and (5) rules out $x = 0$. Consequently $x > 0$ and $|q| \geq 2$. Since $q \equiv 1 \pmod 4$, in fact $|q| \geq 3$. The retained request q^{y+1} now gives $y + 1 \geq 0$. An odd nonzero multiple of $F > 1$ cannot lie in $[-1, 0]$, so $y \geq F$.

If $u_0 \leq -1$, (4) would imply $x \leq F - y \leq 0$. Therefore $u_0 \geq 0$ and $x \geq F + 1$. Equation (5) then gives $u_t \geq 1$ and the second lower bound in (7). Finally $|BH| \geq 2$, $x \geq 3$, and $H \geq 2$ give the remaining bounds. □

Within this particular unconditional numerator-base argument, modulus four is minimal. A progression of modulus one, two, or three cannot avoid all three consecutive forbidden numerator bases $-1, 0, 1$. The progression $q + 1 \equiv 2 \pmod 4$ does avoid them. This is a small, precisely scoped optimality observation, not a lower bound on general Diophantine representations.

5 A smaller frame and an affine proof horizon

The sparser comparison profile retains $t = 16 + h^2$ and $F = [2(t + 1)]^t$. The main profiles instead use

$$t = 8h, \qquad d = t + 2, \qquad F = d^t, \qquad h \in \mathbb{Z}. \tag{8}$$

Three compatible changes keep the resulting coefficients small: bound all affine atoms by one coordinate frame, remove a repeated horizon term from the orientation error pool, and use the smaller growth base $t + 2$. The proof below establishes orientation before using any sign of the horizon.

5.1 A coordinate frame for affine atoms

Lemma 5.1 (Integer affine majorants)**.** *Let $L(\boldsymbol{u}) = c + \sum_i a_i u_i$ have integer coefficients. Define*

$$b(L) = |c| + \sum_i \left\lfloor a_i^2/4 \right\rfloor .$$

Then, for all integer $\boldsymbol{u}$,

$$|L(\boldsymbol{u})| \le \sum_{a_i \neq 0} u_i^2 + b(L).$$

Proof. For integer a, u, both au and $-au$ are at most $u^2 + \lfloor a^2/4 \rfloor$, by maximizing $au - u^2$ over the integers. Use the triangle inequality and sum these bounds after collecting coefficients of the same coordinate. □

Resolve only the new affine source identities for t, d, R, and q. The complete source has 37 distinct requested affine atoms, using 33 underlying coordinates. The supplied `frame_certificate.json` lists every atom, every substitution, and its exact $b(L)$. Their maximum is 18, attained by $d = 8h + 2$. We replace the old frame by

$$W = 19 + \sum_{u \in \mathcal{U}} u^2, \tag{9}$$

where $\mathcal{U}$ is precisely that set of 33 coordinates. Therefore $W \ge 19$ and every actual base, exponent, and output atom has absolute value strictly less than W. The frame coordinates are semantic inputs and output registers, all chosen before the private Pell witnesses; there is no dependence on A or a Pell ordinate.

For examples, the frame costs of $t = 8h$, $t + 1$, $F + t$, $q = 4z + 1$, $q + 1$, and $d = 8h + 2$ are respectively $16, 17, 16, 5, 6, 18$. An ordinary register u costs zero; a translated register $u + 1$ costs one. The lemma covers all 37 atoms uniformly. Unlike the former frame, (9) does not assert that each affine atom's *square* is below W; the following argument needs only the established absolute-value bounds.

5.2 An invertible change of orientation errors

Let $e_t = Y_t^{\text{proposed}} - t$ and $e_{F+t} = Y_{F+t}^{\text{proposed}} - (F + t)$ denote two of the nominal ordinate errors. Keep e_t and replace the second by

$$e_{F+t} - e_t = Y_{F+t}^{\text{proposed}} - Y_t^{\text{proposed}} - F. \tag{10}$$

Keep every other ordinate error and every abscissa error. Squaring and pooling this new list enforces exactly the same zero-error conditions, because the change of error basis is invertible over $\mathbb{Z}$. It removes one repeated copy of the horizon from the quadratic pool.

Here is the unbounded soundness estimate; positive definiteness alone would not suffice to authenticate a congruence. The retained signed anchor and its cap first bound every classified index by the absolute nominal anchor index, hence strictly below W. Each ordinary small ordinate error has absolute value less than $2W$, and each abscissa error has absolute value at most two. In the new difference term, the two classified signed indices and the separate atom F each have absolute value below W, so its absolute value is less than $3W$. With at most fifteen pairs, the reduced nonnegative error obeys

$$E < 65W^2 + 60 < 4W^3 < |A - 1| \qquad (W \ge 19).$$

The orientation row makes $A - 1$ divide E, so $E = 0$. Invertibility of (10) recovers each original ordinate error as zero as well, and both coordinates of every Pell pair are oriented.

The exceptional anchor branch $|C_{\mathrm{Pell}}| = 1$ is handled before asserting the anchor-index identification. Its squared cap forces nominal anchor zero and every other principal ordinate zero. Its classified anchor index is one, and every other index is zero, so the same size bound applies. Its anchor error is positive and is unchanged by (10); therefore $E > 0$, a contradiction.

The odd frame norm still classifies at center $W + 1$ with a positive index divisible by W. It gives $|A| \geq X_W(W + 1) > W^W$ and $|A - 1| > 4W^3$. The power-modulus estimate $|2An - n^2 - 1| > W^W + W$ for $|n| \geq 2$ follows as before. Hence the full signed power interface is available before the horizon has a sign. For constructive completeness, canonical errors and their differences are divisible by $A - 1$, so the changed pooled row has its integer quotient. The semantic-anchor cap and its dominance construction are unchanged.

5.3 Recovering the horizon and the required growth

In (8), $d = 8h + 2$ has absolute value at least two for all integer h. Lemma 4.1 forces $t \geq 0$ and gives $F = d^t$. If $t = 0$, then $F = 1$ and $y = 2v + 1$ is odd. Subtracting the two beta endpoints gives

$$1 = (y + 1)(u_t - u_0),$$

which is impossible because the right side is even. No sign of y and no division by an endpoint modulus is used here. Thus t is a positive multiple of eight and $t \geq 8$.

Now $d = t + 2$ is positive and even, and

$$F \geq 10^8, \qquad F > (t + 1)^t, \qquad v_2(F) \geq t.$$

The selector inequality $F > L_t^2$ follows from Corollary 3.2 whenever $t \geq 16$. The only additional positive multiple of eight is $t = 8$, for which

$$L_8 = 840, \qquad L_8^2 = 705\,600 < 10^8 = F.$$

Thus the first-digit and CRT proofs apply at every possible solution. Proposition 4.2 gives $x \geq 2 + (t + 1)F \geq 900\,000\,002 > 2^{29}$. This is already more than sufficient for the local-rule majorant used below; the older 2^{32} threshold was unnecessarily strong.

Finally, a single explicitly certified integral residual-row subtraction cancels the duplicated leading horizon square in the growth-modulus and pooled-error rows. This last change preserves their common zero set by an elementary inverse. It is an algebraic coefficient reduction, not another semantic assumption. It restores the maximum residual coefficient to 64 and the collected quartic height to 7,968.

6 Authenticating the first digit with a signed radix

At the stage of Proposition 4.2, B and q could still be negative. Write $Q = |q|$ and $c_j = \binom{F+t}{j}$. For $0 \leq j \leq t$,

$$c_j \leq (F^2 + F)^t \leq 2^{t(F-1)}.$$

The last elementary inequality holds for $F \geq 7$. Combining it with $Q \geq 2^x$ and $x \geq 2 + (t + 1)F$ gives

$$\frac{\max_{j \leq t} c_j}{Q - 1} < \frac{1}{32}, \qquad c_t < Q/32.$$

The retained common denominator is $M = q^x$. Each digit denominator $D_i = q^{k_i}$ divides M, and the source uses its integer quotient v_i together with normalized remainder ρ_i. Its one shared bound is

$$M^2 = 1 + 4\sum_{i=1}^{11} \rho_i^2 + \alpha^2 + \beta^2 + \gamma(\gamma+1). \tag{11}$$

All final coordinates are integers, so the last three terms are nonnegative. Thus $M \neq 0$ and $|\rho_i| < |M|/2$.

The scaled digit row implies $v_i \mid \rho_i$: every other term is divisible by v_i, since $M = v_i D_i$. Accordingly $r_i = \rho_i / v_i$ is an *integer*, and $|r_i| < |D_i|/2$. This integrality point prevents a rational-relaxation gap in the normalized-remainder argument.

For the first digit the signed low polynomial is

$$r_* = \sum_{j=0}^{t-1} c_j q^j.$$

Its absolute value is less than $Q^t/32$, and it is congruent to $(q+1)^{F+t}$ modulo q^t. The certified remainder has the same congruence and absolute value less than $Q^t/2$. Their difference has absolute value less than Q^t, so it is zero. The exact digit row now gives $B \equiv c_t \pmod q$. Since

$$|B - c_t| < Q/2 + Q/32 < Q,$$

we obtain $B = c_t = \binom{F+t}{t} > 0$. Only now does $q = (BH)^x$ imply $q > 0$. No earlier step assumed a positive radix.

7 From the remaining digits to a finite refutation

Once $B, q > 0$, the predecessor's remaining range proof applies. For an ordinary nonnegative upper representative a and $0 \le j \le k \le x$,

$$\binom{a}{j} \le \max(1, a)^k \le H^x.$$

This includes zero representatives. The retained derived-label majorant has exponent $\ell = 2^{22}$. The endpoint bound and $F \ge 10^8 > 2^{22}$ give $\ell(t+1) < x$, so that label's required coefficients obey the same bound. Since $q = B^x H^x$, all eleven true normalized remainders are less than $M/32$ and are recovered uniquely by (11).

For the local numerical rule polynomial the retained majorant is $\mathcal{L} < \mathcal{R}^E$ with $\mathcal{R} = 2 + t + x + y + F$ and $E = 2^{23}$. The bounds $y \ge F$, $x \ge 2 + (t+1)y$, and $t \ge 8$ imply $\mathcal{R} < 2x$. For integers $x \ge 2^{29}$,

$$(2x)^E < 2^x.$$

At $x = 2^{29}$, its logarithmic inequality is $30 \cdot 2^{23} < 2^{29}$. One then observes $(1 + 1/x)^E < 1/(1 - E/x) < 2$ to induct. Our endpoint bound gives $x \ge 900\,000\,002 > 2^{29}$. Thus every required local value is strictly below the prime-power precision 2^x.

The selector facts following Corollary 3.2 now recover every bounded row. To see the essential argument, a selector $p^e \| B$ with $p^e > t$ identifies its row by the raw congruence $B \mid C + F + 1$. The binomial range conditions recover all bounded columns modulo p^{xe}. The local polynomial is divisible by $p^{xe} \ge 2^x$ and lies in $[0, 2^x)$, so it is zero.

The algebraic boundary $y = F$ is still possible in the affine beta formula and must be excluded. The first endpoint would give initial accumulator zero modulo $y + 1$, since $F + 1 = y + 1$. Every local update has the form

$$S_{i+1}(1 + v_i F) = S_i(1 + p_i F)(1 + q_i F),$$

with positive label weights. Zero propagates through all rows, contrary to the final accumulator one. Hence $y > F$, and the initial accumulator is the intended $F + 1$.

Multiplying the updates gives an equality of products of weights $1 + jF$. The conclusion side has at most t nonunit factors; the other side, including the initial target, has at most $2t + 1$. If the counts differ, their smaller value r is at most t. The larger product is at least $(F + 1)^{r+1}$ and the smaller at most $t^r(F + 1)^r$. This contradicts $F + 1 > t^t$. Thus both counts are equal and at most t. Replace F by an indeterminate. Every coefficient sum is at most $(t + 1)^t < F$. A nonzero difference polynomial cannot vanish at F: its lowest nonzero coefficient would have to be divisible by F despite having smaller absolute value. Unique factorization therefore recovers the label multiplicities.

For the single-conclusion flow step, take the closure of truth under the finitely many certified rule instances. If the target were outside this closure, each rule with conclusion outside would consume at least one premise outside. The total number of outside conclusions minus outside premises would be nonpositive, whereas the target balance makes it one. This contradiction yields a finite C222 derivation. The named logical calibration completes the soundness direction of Theorem 1.1.

8 Constructive completeness

Start with a finite calibrated C222 refutation. Pad with truth steps to a sufficiently large permitted horizon: $16 + h^2$ for the square-horizon profile, or a positive multiple of 8 for the affine profiles. The horizon is chosen beyond the proof size and its finite local parameters. Define F by the corresponding growth request.

Construct the pending-multiset accumulators using the weights $1 + jF$. Let O be the odd part of $t!$. Choose a sufficiently large positive odd multiple w of O and set

$$y = Fw, \qquad v = (w - 1)/2.$$

Every selected growth base is even, so $v_2(F) \geq t > v_2(t!)$. Consequently y is an even multiple of $t!$. It can be chosen larger than every accumulator and larger than F. The beta moduli $1 + (i + 1)y$, $0 \leq i \leq t$, are pairwise coprime and odd. CRT supplies a code x; adding their odd product permits an arbitrarily large *even* code while preserving all residues.

Set $B = \binom{F+t}{t}$ and $N = B^x$. For $1 \leq i \leq t$, $v_2(F) > v_2(i)$, so $v_2(F + i) = v_2(i)$. The numerator and denominator in the binomial therefore have equal 2-adic valuation, and B is odd. The generalized binomial CRT theorem supplies all representative classes modulo N^2.

Choose the distinguished schema representative a odd and all other actual representatives even. The odd period N^2 permits these choices. Choose a in $[2N^2, 4N^2)$ and the others positive and at most $2N^2$. The unchanged constructor identities make the semantic anchor even and at least $2a^2$, so it exceeds twice every other required index. The front index $F + t$ is at most B for $t \geq 2$, and the other front indices are smaller than N^2. The bounds used in the retained anchor construction therefore remain valid.

Now $H = 2 + C + \sum A_i$ is odd. Since B, H are odd and x is even, $q = (BH)^x$ is an odd square; in particular $q \equiv 1 \pmod 4$. Thus $z = (q - 1)/4$ is an integer. This uses precisely the predecessor's parity freedom; no new modulus-three condition is needed.

Choose the canonical shared Pell witnesses and normalized remainders. All eleven true remainders satisfy $0 \leq \rho_i < M/32$, so $4 \sum \rho_i^2 < M^2$. The nonnegative integer gap in (11) is representable as $\alpha^2 + \beta^2 + \gamma(\gamma + 1)$.

For example, Legendre's three-square theorem applied to $4g + 1$ gives this representation of any $g \geq 0$: exactly one square is odd modulo four. All unchanged private certificates are supplied by the retained constructive Pell argument. This constructs an integer zero of the new raw source, and the exact algebraic projections give zeros of the delivered quartics.

Notice that F is chosen before the large beta multiplier w. Identifying y with $F + 1$ would destroy this freedom. An accumulator with two pending nonunit labels already has value at least $(F + 1)^2$, which cannot fit below $y = F + 1$. That tempting further simplification was rejected for this precise completeness obstruction.

9 From the source certificate to the literal quartics

The semantic changes preserve *existence* by the two refutation translations above. They are not asserted to give a pointwise bijection with every zero of the predecessor polynomial. In contrast, the later finite algebraic steps give exact existential projections with specified integral restorations; invertible row and coordinate changes give bijections. They neither add nor lose the existence of integer zeros.

The supplied producer records the raw edits, inherited Horn factorizations, monic coordinate eliminations, integral translations, affine residual exposures, and reversible residual row operations. Every eliminated coordinate has an explicit integer-polynomial restoration. Each elementary row operation is invertible over the integers. Finally a sum of real squares vanishes exactly when every quadratic residual vanishes.

The final sparse serialization uses a term $[c, [i, j, \ldots]]$ for $cx_ix_j \cdots$. The empty index list denotes a constant; repeated indices denote repeated factors. Thus these files specify actual ordinary integer polynomials independently of any informal semantic register name.

9.1 Verification scope and reproduction

The frozen finite checks passed in ordinary and optimized Python modes. The ordinary run also regenerated all 37 generated JSON files in a fresh temporary directory and compared them byte for byte with the archive. No external Python package is required for the polynomial checks.

- The source checker verifies every new source edit, the complete 37-atom/33-coordinate frame ledger, 18 inherited Horn identities, and 205 or 206 monic projections, depending on the profile.
- Each final endpoint has six coordinate shears, five affine-exposure operations, and three or four affine projections. Every final residual row operation is checked exactly over the integers.
- The independent collector recomputed all six delivered quartics (two intermediate systems and four final endpoints), including every collected coefficient and every claimed metric.
- The source/map checker rejected 16 deliberate corruptions across its two profiles. The separate polynomial collector rejected another 18 corrupted residual, coefficient, and metric examples.
- The literal LaTeX appendices and plain-text residual lists are regenerated from the frozen equations; the export check requires exact equality with the supplied text.

From the unpacked package root, run:

```
python3 -B verify.py
python3 -B verify.py --rebuild
```

```
python3 -B -O code/verify_endpoints.py
sh build_report.sh
```

The first command checks the archive manifest and all finite certificates. The second additionally rebuilds the 37 JSON files. The third verifies that the checks do not depend on Python assertions. The last command requires a standard pdfLaTeX installation and rebuilds this report.

The independent root collector imports no producer or inherited compiler. It recomputes every coefficient of every delivered sum of squares, checks all reported dimensions, degrees, heights and supports, and requires corrupted coefficient, residual, and metric examples to fail. A separate checker validates the raw source changes and the complete finite algebraic transformations. The executable receipts state what was actually run.

These exact finite checks establish that the delivered data implement the specified polynomial transformations. The unbounded Pell, prime-power, CRT, sign, and flow arguments above establish the mathematical meaning. Neither kind of evidence is presented as a substitute for the other. The full transitive Lean/Rocq development was not rebuilt, and this packet does not contain a newly checked encoded object-language ZFC derivation.

10 Further research questions

1. Can the small square-frame estimate support further affine parameters in the private Pell certificates? Each proposal must exclude zero or unit bases before invoking signed exponent recovery, and must retain the anchor's squared divisibility.

2. Can the common remainder norm reuse an existing universal ternary representation without restricting its integer gap? Replacing a three-variable nonnegativity certificate by a two-square demand would need a new completeness theorem; size alone is insufficient.

3. Can a different finite proof calculus reduce the maximal pending multiset width and thereby lower the beta scale? The failed $y = F + 1$ attempt identifies the concrete obstruction that such a calculus would need to remove.

4. Can exact lattice changes reduce the support or lower the coefficient height below 7,968 while retaining the affine horizon's smaller dimension? Higher-degree ideal consequences or different source coordinates would be a substantive next step.

5. Can the new raw source and its sign-recovery order be added to the repository's paused proof-assistant development as a small independent theorem? The best first target is the combined bootstrap, before attempting the full polynomial compilation.

Attribution and integration

The new work is the compatible affine bootstrap and linear selector composition, its proofs, literal endpoints, and finite certificates. The numerical calculus, logical calibration, factorial-free CRT architecture, shared Pell frame, normalized remainder construction, cofinal parity choices, and earlier affine and row reductions are inherited and credited. The package retains the predecessor and its source notices. Independent review here means separate AI-assisted mathematical and algebraic reviews, not external human refereeing. No upstream branch, issue, pull request, or repository file was modified.

References

[1] V. Reshetnikov, *ProveIt*, pinned repository snapshot `7a8da414376a80b90ecf799f37ceb26fb9ac674e`. https://github.com/VladimirReshetnikov/ProveIt.

[2] ProveIt incoming research packet, *Removing factorial authenticity from Diophantine consistency*, 25 October 2026. Retained verbatim in `inherited_factorial_free/`.

[3] ProveIt incoming research packet, *Shared factorial errors: explicit Diophantine equations for Con(ZFC)*, 25 October 2026. Comparison identity recorded in `PROVENANCE.json`.

[4] B. Farhi, *An identity involving the least common multiple of binomial coefficients and its application*, American Mathematical Monthly 116 (2009), 836–839. https://arxiv.org/abs/0906.2295.

[5] H.-L. Chan and M. Norrish, *Proof Pearl: Bounding Least Common Multiples with Triangles*, Journal of Automated Reasoning 62 (2019), 171–192. https://doi.org/10.1007/s10817-017-9438-0.

[6] J. Bayer, M. David, M. Haßler, Y. Matiyasevich and D. Schleicher, *Diophantine Equations over $\mathbb{Z}$: Universal Bounds and Parallel Formalization*, arXiv:2506.20909, version 3, 22 October 2026. https://arxiv.org/abs/2506.20909v3.

A The complete main residual list

Every displayed coordinate is an unrestricted integer. This appendix defines all residuals of (1); no omitted family or implicit coefficient parameter is required. The sparser comparison list is supplied in its data directory.

$$
\begin{aligned}
R_0 =& -x_1 - 8x_1x_{212} + x_6 + 8x_{212}\\
R_1 =& -x_1x_4 - x_2 + x_5x_6\\
R_2 =& -1 + x_0 - x_5x_6\\
R_3 =& 1 + x_2 - x_3x_{17} + x_{15}\\
R_4 =& -1 - x_7^2 - 2x_7x_8 - x_8^2 + x_{18}\\
R_5 =& -10x_7 + x_7^2 - 6x_7x_8 - 8x_7x_{18} - 20x_8\\
&- 11x_8^2 - 16x_8x_{18} - 15x_{18} - 4x_{18}^2 + x_{19}\\
R_6 =& -2 + 2x_7 - x_7^2 - 4x_7x_{19} - 4x_{19}\\
&- 4x_{19}^2 + x_{20}\\
R_7 =& 1 + 6x_7 - 36x_7^2 - 24x_7x_{20} + 2x_{20}\\
&- 4x_{20}^2 + x_{21}\\
R_8 =& -2 - x_8 - x_8^2 - 2x_8x_9 - x_9\\
&- x_9^2 + x_{22}\\
R_9 =& -8 + 7x_7 - x_7^2 + 4x_7x_9 - 4x_7x_{22}\\
&- 4x_9 + 4x_{22} + x_{23} - x_{45}\\
R_{10} =& -1 + x_7 - x_7^2 - 2x_7x_9 + x_9\\
&- x_9^2 + x_{24}\\
R_{11} =& 23 + 16x_7 - 8x_7x_8 - 8x_7x_9 - 16x_7x_{18}\\
&- 8x_7x_{24} + 19x_8 + 9x_8^2 - 36x_8x_9 - 12x_8x_{24}\\
&+ 16x_9 + 8x_9^2 - 24x_9x_{18} - 8x_{18}x_{24} - x_{19}\\
&+ x_{25} - x_{30}\\
R_{12} =& -4x_{23}^2 - 8x_{23}x_{25} - 4x_{25}^2 + x_{26}\\
R_{13} =& -2 - 8x_7 + 8x_7x_{22} + 8x_7x_{24} - 8x_8\\
&+ 8x_8x_{22} + 8x_8x_{24} - 4x_{18} + 8x_{22} - 4x_{22}^2\\
&- 8x_{22}x_{24} + 8x_{24} - 4x_{24}^2 + x_{27}\\
R_{14} =& 6 + 4x_7 + 32x_7x_8 + 16x_7x_{18} + 16x_7x_{24}\\
&+ 8x_7x_{27} + 24x_8 - 32x_8x_{24} - 16x_8x_{27} + 12x_{18}\\
&- 16x_{18}x_{24} - 8x_{18}x_{27} - x_{19} + 8x_{24} - 16x_{24}^2\\
&- 16x_{24}x_{27} + 4x_{27} - 4x_{27}^2 + x_{28}\\
R_{15} =& 22 - 5x_7 - x_7^2 - 8x_7x_8 - 12x_7x_9\\
&- 4x_7x_{22} - 8x_8 - 16x_8x_9 - 16x_8x_{22} - 20x_9\\
&- 20x_9^2 - 24x_9x_{22} - 28x_{22} - 4x_{22}^2 + x_{29}\\
R_{16} =& 19 - 25x_7 - 7x_7^2 - 8x_7x_8 - 30x_7x_9\\
&- 16x_7x_{24} + 3x_8 + 5x_8^2 - 4x_8x_{24} - 27x_9\\
&- 21x_9^2 - 24x_9x_{24} - 6x_{22} - 17x_{24} - 4x_{24}^2\\
&+ x_{30}\\
R_{17} =& 2 - 4x_{29}^2 - 8x_{29}x_{30} - 4x_{30}^2 + x_{31}\\
R_{18} =& -19x_7 + 8x_7^2 + 8x_7x_{18} + 8x_7x_{22} - 46x_8\\
&- 24x_8^2 + 24x_8x_{18} + 24x_8x_{22} - 2x_9 + 4x_{18}\\
&- 4x_{18}^2 - 8x_{18}x_{22} + 14x_{22} - 4x_{22}^2 - x_{24}\\
&+ x_{32}\\
R_{19} =& -3 - 6x_7 - 4x_9 - 2x_{24} - 4x_{32}\\
&- 4x_{32}^2 + x_{33}
\end{aligned}
$$

$$
\begin{aligned}
R_{20} &= 8 + 42x_7 + 7x_7^2 + 28x_7x_{19} + 48x_8 \\
&\quad - 32x_8^2 - 32x_8x_{18} + 64x_8x_{19} - 40x_{18}^2 + 32x_{18}x_{19} \\
&\quad + 14x_{19} - 4x_{19}^2 + x_{34} + 2x_{37} \\
R_{21} &= 1 + 6x_7 - 36x_7^2 - 24x_7x_{34} + 2x_{34} \\
&\quad - 4x_{34}^2 + x_{35} \\
R_{22} &= -x_8^2 - 4x_8x_{35} - 4x_{35}^2 + x_{36} \\
R_{23} &= -6 - 2x_7 - 4x_8 - 14x_{18} - 16x_{18}^2 \\
&\quad + x_{37} \\
R_{24} &= 24 + 21x_8 - 21x_8^2 - 18x_8x_9 - 24x_8x_{22} \\
&\quad - 24x_8x_{37} + 3x_9 - 9x_9^2 - 16x_9x_{37} + 3x_{22} \\
&\quad - 12x_{22}^2 - 8x_{22}x_{37} + 4x_{37} - 4x_{37}^2 + x_{38} \\
&\quad + 2x_{45} \\
R_{25} &= 16 - 10x_7 - 20x_7^2 - 16x_7x_9 - 24x_7x_{24} \\
&\quad - 24x_7x_{38} - 12x_9 - 16x_9x_{24} - 16x_9x_{38} - 14x_{24} \\
&\quad - 4x_{24}^2 - 8x_{24}x_{38} + 4x_{38} - 4x_{38}^2 + x_{39} \\
R_{26} &= -x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{40} \\
R_{27} &= 2 + x_7^2 - 4x_7x_{40} + 2x_9 - 2x_9^2 \\
&\quad - 8x_9x_{40} - 2x_{24} + 4x_{40} - 4x_{40}^2 + x_{41} \\
R_{28} &= -31 - 16x_9 - 16x_9x_{22} + 16x_9x_{24} - 8x_{22} \\
&\quad + 12x_{22}^2 - 8x_{22}x_{24} + 24x_{24} - 4x_{24}^2 + x_{42} \\
&\quad - 4x_{45} \\
R_{29} &= -4x_9 - 16x_{24}^2 - 32x_{24}x_{40} - 16x_{24}x_{41} - 16x_{24}x_{42} \\
&\quad - 4x_{40} - 16x_{40}^2 - 16x_{40}x_{41} - 16x_{40}x_{42} - 2x_{41} \\
&\quad - 4x_{41}^2 - 8x_{41}x_{42} - 4x_{42}^2 + x_{43} \\
R_{30} &= -16x_7^2 + 8x_7x_9 + 32x_7x_{24} + 16x_7x_{27} + x_9 \\
&\quad - x_9^2 - 8x_9x_{24} - 4x_9x_{27} - 16x_{24}^2 - 16x_{24}x_{27} \\
&\quad - 4x_{27}^2 + x_{44} \\
R_{31} &= -1 - 8x_9 - 4x_9^2 + 8x_9x_{22} + 8x_{22} \\
&\quad - 4x_{22}^2 + x_{45} \\
R_{32} &= -1 - 9x_9^2 + 24x_9x_{22} + 12x_9x_{45} - 16x_{22}^2 \\
&\quad - 16x_{22}x_{45} - 4x_{45}^2 + x_{46} \\
R_{33} &= 1 + x_8 - x_8^2 - 4x_8x_9 - 4x_8x_{46} \\
&\quad - 4x_9^2 - 8x_9x_{46} - 4x_{46}^2 + x_{47} \\
R_{34} &= -1 - 4x_{47} - 4x_{47}^2 + x_{48} \\
R_{35} &= -1 - x_9^2 - 8x_9x_{47} - 4x_9x_{48} - 16x_{47}^2 \\
&\quad - 16x_{47}x_{48} - 4x_{48}^2 + x_{49} \\
R_{36} &= 2 + x_8 - x_8^2 - 4x_8x_9 - 4x_8x_{49} \\
&\quad - 4x_9^2 - 8x_9x_{49} - 4x_{49}^2 + x_{50} \\
R_{37} &= -4x_{44}^2 - 8x_{44}x_{50} - 4x_{50}^2 + x_{51} \\
R_{38} &= -9 + 7x_7 - x_7^2 - 8x_7x_{44} - 4x_7x_{51} \\
&\quad + 24x_{44} - 16x_{44}^2 - 16x_{44}x_{51} + 12x_{51} - 4x_{51}^2 \\
&\quad + x_{52} \\
R_{39} &= -1 - 4x_{52} - 4x_{52}^2 + x_{53} \\
R_{40} &= -x_9^2 - 8x_9x_{52} - 4x_9x_{53} - 16x_{52}^2 - 16x_{52}x_{53} \\
&\quad - 4x_{53}^2 + x_{54} \\
R_{41} &= -x_7 - x_7^2 - 4x_7x_9 - 4x_7x_{54} - 4x_9 \\
&\quad - 4x_9^2 - 8x_9x_{54} - 4x_{54} - 4x_{54}^2 + x_{55}
\end{aligned}
$$

$$
\begin{aligned}
R_{42} &= -10 - 12x_{55} - 4x_{55}^2 + x_{56} \\
R_{43} &= 2 + 4x_9 + 16x_9x_{55} + 8x_9x_{56} - 4x_{22} \\
&\quad - 16x_{22}x_{55} - 8x_{22}x_{56} - x_{45} + 8x_{55} - 16x_{55}^2 \\
&\quad - 16x_{55}x_{56} + 4x_{56} - 4x_{56}^2 + x_{57} \\
R_{44} &= x_8 - x_8^2 + 8x_8x_9 - 8x_8x_{22} - 4x_8x_{57} \\
&\quad - 16x_9^2 + 32x_9x_{22} + 16x_9x_{57} - 16x_{22}^2 - 16x_{22}x_{57} \\
&\quad - 4x_{57}^2 + x_{58} \\
R_{45} &= -10 - 12x_{50} - 4x_{50}^2 + x_{59} \\
R_{46} &= 1 - 16x_{50}^2 - 16x_{50}x_{58} - 16x_{50}x_{59} - 4x_{58}^2 \\
&\quad - 8x_{58}x_{59} - 4x_{59}^2 + x_{60} \\
R_{47} &= x_9 - x_9^2 - 8x_9x_{58} - 4x_9x_{60} - 16x_{58}^2 \\
&\quad - 16x_{58}x_{60} - 4x_{60}^2 + x_{61} \\
R_{48} &= -6x_7 - 3x_7^2 - 18x_7x_8 - 16x_8 - 27x_8^2 \\
&\quad + 4x_8x_9 + 3x_9 + 3x_9^2 - 4x_9x_{18} - 19x_{18} \\
&\quad - 8x_{18}^2 + x_{19} + x_{62} - 4x_{75} \\
R_{49} &= 1 - 8x_7 - 4x_7x_9 + 8x_7x_{62} - 16x_8 \\
&\quad - 8x_8x_9 + 16x_8x_{62} + x_9 + x_9^2 + 4x_9x_{18} \\
&\quad + 8x_{18} - 8x_{18}x_{62} + 3x_{62} - 4x_{62}^2 + x_{63} \\
R_{50} &= -2 - 12x_7 - 16x_7x_9 + 16x_7x_{61} + 16x_7x_{63} \\
&\quad - 24x_8 - 32x_8x_9 + 32x_8x_{61} + 32x_8x_{63} + 4x_9 \\
&\quad + 4x_9^2 + 16x_9x_{18} + 12x_{18} - 16x_{18}x_{61} - 16x_{18}x_{63} \\
&\quad - 4x_{61}^2 - 8x_{61}x_{63} - 4x_{62} - 2x_{63} - 4x_{63}^2 \\
&\quad + x_{64} \\
R_{51} &= -2 - 4x_{64} - 4x_{64}^2 + x_{65} \\
R_{52} &= -x_2x_{12} + x_{66} \\
R_{53} &= x_{11} - x_{12} - 2x_{21}x_{66} - x_{67} \\
R_{54} &= x_{11} - x_{12} - 4x_{23}x_{66} - 2x_{26}x_{66} - x_{68} \\
R_{55} &= -2x_{28}x_{66} + 2x_{43}x_{66} - 2x_{66} - x_{69} + x_{74} \\
R_{56} &= x_{11} - x_{12} - 4x_{29}x_{66} - 2x_{31}x_{66} - x_{70} \\
R_{57} &= x_{11} - x_{12} - 2x_{33}x_{66} + 6x_{66} - x_{71} \\
R_{58} &= x_{11} - x_{12} - 4x_{35}x_{66} - 2x_{36}x_{66} - x_{72} \\
R_{59} &= x_{11} - x_{12} - 2x_{39}x_{66} - x_{73} \\
R_{60} &= x_{11} - x_{12} - 2x_{43}x_{66} - 4x_{66} - x_{74} \\
R_{61} &= -x_7x_8 + x_7x_9 + x_8 + x_8x_9 - x_9 \\
&\quad - x_9^2 + x_{75} \\
R_{62} &= 1 + 2x_{10} + x_{10}^2 - x_{75}^2 + x_{76} \\
R_{63} &= -x_{76}^2 + x_{77} \\
R_{64} &= -4x_{64}x_{66} - 2x_{65}x_{66} + 2x_{66} + x_{78} \\
R_{65} &= -x_{11}^2 + 2x_{11}x_{12} + 2x_{11}x_{78} - x_{12}^2 - 2x_{12}x_{78} \\
&\quad - x_{78}^2 + x_{79} \\
R_{66} &= -1 - x_2x_8 + x_{80} \\
R_{67} &= -x_{12}x_{80} + x_{82}x_{83} + x_{84} \\
R_{68} &= -1 + x_2 - x_2x_7 + x_{81} \\
R_{69} &= -1 + 2x_2 + 2x_2x_7 + 4x_2x_8 - 2x_2x_{18} \\
&\quad + x_{82} \\
R_{70} &= -x_{11}x_{81} + x_{83} \\
R_{71} &= -6x_2x_{12} + x_{11}x_{80} + 12x_{12} - 8x_{12}x_{80} - 4x_{12}x_{81} \\
&\quad - x_{12}x_{82} + x_{85} \\
R_{72} &= -x_{11}x_{67} + x_{12}x_{67} + x_{86}
\end{aligned}
$$

$$
\begin{aligned}
R_{73} &= -x_{68}x_{86} + x_{87} \\
R_{74} &= -x_{69}x_{87} + x_{88} \\
R_{75} &= -x_{70}x_{88} + x_{89} \\
R_{76} &= -x_{71}x_{89} + x_{90} \\
R_{77} &= -x_{72}x_{90} + x_{91} \\
R_{78} &= -x_{73}x_{91} + x_{92} \\
R_{79} &= -x_{74}x_{92} + x_{93} \\
R_{80} &= -x_{84}x_{93} + x_{94} \\
R_{81} &= -x_{85}x_{94} + x_{95} \\
R_{82} &= -x_{95}^2 + x_{96} \\
R_{83} &= -x_1 - x_1x_{15} + x_{15} + x_{97} \\
R_{84} &= -x_{13}x_{97} + x_{98} \\
R_{85} &= -2x_0^2 + 2x_0x_{11} + 2x_0x_{12} + 2x_0x_{98} + 2x_0x_{99} \\
&\quad - x_{11}^2 - 2x_{11}x_{98} - x_{12}^2 - 2x_{12}x_{99} + x_{16}x_{101} \\
&\quad - x_{77}x_{96} - x_{79}x_{96} - x_{98}^2 - x_{99}^2 \\
R_{86} &= -x_1x_{14} + x_{14} - x_{14}x_{97} + x_{99} \\
R_{87} &= -x_7 - x_8 - 4x_{18} - 2x_{21} - 4x_{23} \\
&\quad - 2x_{26} - 2x_{28} - 4x_{29} - 2x_{31} - 2x_{33} \\
&\quad - 4x_{35} - 2x_{36} - 2x_{39} - 2x_{43} - 4x_{64} \\
&\quad - 2x_{65} + x_{100} \\
R_{88} &= -x_{104} - 4x_{104}x_{236} + x_{106} \\
R_{89} &= -x_3x_{128} + x_{102}x_{227} - x_{105} - x_{106}x_{128} \\
R_{90} &= x_{107}x_{228} - x_{109}x_{131} - x_{110} \\
R_{91} &= x_{111}x_{228} - x_{112}x_{131} - x_{113} \\
R_{92} &= x_{114}x_{228} - x_{115}x_{131} - x_{116} \\
R_{93} &= x_{117}x_{228} - x_{118}x_{131} - x_{119} \\
R_{94} &= x_{120}x_{229} - x_{122}x_{131} - x_{123} \\
R_{95} &= x_{124}x_{229} - x_{125}x_{131} - x_{126} \\
R_{96} &= -x_{16}x_{128} + x_{131} \\
R_{97} &= x_{127} - x_{129}x_{131} - x_{130} \\
R_{98} &= -x_{131}x_{133} + x_{132} - x_{134} \\
R_{99} &= -x_{131}x_{136} + x_{135}x_{227} - x_{137} \\
R_{100} &= -x_{131}x_{139} + x_{138}x_{228} - x_{140} \\
R_{101} &= x_{141} - 2x_{141}x_{143} + x_{144} \\
R_{102} &= -x_{141}x_{144} - x_{144} + x_{145} \\
R_{103} &= x_{144}^2 - x_{145}^2 + x_{165} \\
R_{104} &= -x_{142}x_{147} - x_{147} + x_{148} \\
R_{105} &= 2x_{146} + x_{146}^2 + x_{147}^2 - x_{148}^2 \\
R_{106} &= 1 - 4x_{142} - 16x_{142}x_{212} + x_{149} + 16x_{212} \\
&\quad + 64x_{212}^2 \\
R_{107} &= -1 + x_2 + x_{142}x_{147} - x_{146} - x_{147} \\
&\quad - 8x_{147}x_{212} + x_{149}x_{150} \\
R_{108} &= 1 - 2x_3 + x_3^2 - 2x_3x_{142} + x_{151} \\
R_{109} &= -x_3x_{196} + x_{16} + x_{151}x_{152} - x_{195} + x_{197} \\
R_{110} &= -1 + 2x_{142}x_{211} - x_{153} + 2x_{211} - x_{211}^2 \\
R_{111} &= 1 + x_{153}x_{154} - x_{195} - x_{196}x_{211} + x_{197} \\
&\quad + 4x_{236} \\
R_{112} &= -x_{142}x_{156} - x_{156} + x_{157} \\
R_{113} &= 2x_{155} + x_{155}^2 + x_{156}^2 - x_{157}^2 \\
R_{114} &= 1 - 4x_{142} - 8x_{142}x_{236} + x_{158} + 8x_{236}
\end{aligned}
$$

$$
\begin{aligned}
&+ 16x_{236}^2 \\
R_{115} = &-1 + x_{102} - x_{155} - 2x_{156} - 4x_{156}x_{236} \\
&+ x_{157} + x_{158}x_{159} \\
R_{116} = &-1 + x_{103} - 2x_{142}x_{160} - x_{146} - x_{147} \\
&- 4x_{147}x_{236} + x_{148} + x_{158}x_{160} + x_{160} + 8x_{160}x_{236} \\
R_{117} = &1 + x_{142} - x_{142}x_{162} - x_{162} + x_{163} \\
R_{118} = &1 + 2x_{161} + x_{161}^2 - 2x_{162} + x_{162}^2 \\
&- x_{163}^2 \\
R_{119} = &1 + x_{107} + x_{158}x_{164} - x_{161} - 2x_{162} \\
&- 4x_{162}x_{236} + x_{163} + 4x_{236} \\
R_{120} = &-2x_{142} - x_{142}^2 + x_{165} \\
R_{121} = &-1 + x_{108} + x_{142}x_{148} - 2x_{142}x_{166} - x_{146} \\
&- 4x_{146}x_{236} - x_{147}x_{165} - 4x_{148}x_{236} + x_{158}x_{166} + x_{166} \\
&+ 8x_{166}x_{236} - 4x_{236} \\
R_{122} = &-x_{142}x_{168} - x_{168} + x_{169} \\
R_{123} = &2x_{167} + x_{167}^2 + x_{168}^2 - x_{169}^2 \\
R_{124} = &-1 + x_{111} + x_{158}x_{170} - x_{167} - 2x_{168} \\
&- 4x_{168}x_{236} + x_{169} \\
R_{125} = &-x_{142}x_{172} - x_{172} + x_{173} \\
R_{126} = &2x_{171} + x_{171}^2 + x_{172}^2 - x_{173}^2 \\
R_{127} = &-1 + x_{114} + x_{158}x_{174} - x_{171} - 2x_{172} \\
&- 4x_{172}x_{236} + x_{173} \\
R_{128} = &-x_{142}x_{176} - x_{176} + x_{177} \\
R_{129} = &2x_{175} + x_{175}^2 + x_{176}^2 - x_{177}^2 \\
R_{130} = &-1 + x_{117} + x_{158}x_{178} - x_{175} - 2x_{176} \\
&- 4x_{176}x_{236} + x_{177} \\
R_{131} = &-x_{142}x_{180} - x_{180} + x_{181} \\
R_{132} = &2x_{179} + x_{179}^2 + x_{180}^2 - x_{181}^2 \\
R_{133} = &-1 + x_{120} + x_{158}x_{182} - x_{179} - 2x_{180} \\
&- 4x_{180}x_{236} + x_{181} \\
R_{134} = &-x_{142}x_{184} - x_{184} + x_{185} \\
R_{135} = &2x_{183} + x_{183}^2 + x_{184}^2 - x_{185}^2 \\
R_{136} = &-1 + x_{121} - 2x_{142}x_{186} + x_{158}x_{186} - x_{183} \\
&- x_{184} - 4x_{184}x_{236} + x_{185} + x_{186} + 8x_{186}x_{236} \\
R_{137} = &-x_{142}x_{188} - x_{188} + x_{189} \\
R_{138} = &2x_{187} + x_{187}^2 + x_{188}^2 - x_{189}^2 \\
R_{139} = &-1 + x_{124} + x_{158}x_{190} - x_{187} - 2x_{188} \\
&- 4x_{188}x_{236} + x_{189} \\
R_{140} = &-x_{142}x_{192} - x_{192} + x_{193} \\
R_{141} = &2x_{191} + x_{191}^2 + x_{192}^2 - x_{193}^2 \\
R_{142} = &-1 + x_{127} + x_{158}x_{194} - x_{191} - 2x_{192} \\
&- 4x_{192}x_{236} + x_{193} \\
R_{143} = &1 - x_{142}x_{196} - x_{196} + x_{197} \\
R_{144} = &-1 + 2x_{195} + x_{195}^2 + x_{196}^2 - 2x_{197} \\
&- x_{197}^2 \\
R_{145} = &x_{128} - 2x_{142}x_{198} + x_{158}x_{198} - x_{195} - x_{196} \\
&- 4x_{196}x_{236} + x_{197} + x_{198} + 8x_{198}x_{236} \\
R_{146} = &-x_{142}x_{200} - x_{200} + x_{201} \\
R_{147} = &2x_{199} + x_{199}^2 + x_{200}^2 - x_{201}^2 \\
R_{148} = &-1 + x_{132} + x_{158}x_{202} - x_{199} - 2x_{200}
\end{aligned}
$$

$$
\begin{aligned}
&\quad - 4x_{200}x_{236} + x_{201} \\
R_{149} &= - x_{142}x_{204} - x_{204} + x_{205} \\
R_{150} &= 2x_{203} + x_{203}^2 + x_{204}^2 - x_{205}^2 \\
R_{151} &= - 1 + x_{135} + x_{158}x_{206} - x_{203} - 2x_{204} \\
&\quad - 4x_{204}x_{236} + x_{205} \\
R_{152} &= - x_{142}x_{208} - x_{208} + x_{209} \\
R_{153} &= 2x_{207} + x_{207}^2 + x_{208}^2 - x_{209}^2 \\
R_{154} &= - 1 + x_{138} + x_{158}x_{210} - x_{207} - 2x_{208} \\
&\quad - 4x_{208}x_{236} + x_{209} \\
R_{155} &= - 32 - x_0^2 + 2x_1 - x_1^2 - x_2^2 \\
&\quad - x_3^2 + 4x_7 + 2x_7x_8 + 2x_8 + 2x_9 \\
&\quad - x_9^2 + 2x_{10} - x_{10}^2 + 2x_{11} - x_{11}^2 \\
&\quad + 2x_{12} - x_{12}^2 + 2x_{13} - x_{13}^2 + 2x_{14} \\
&\quad - x_{14}^2 + 2x_{15} - x_{15}^2 - x_{16}^2 - x_{18} \\
&\quad + 2x_{100} - x_{100}^2 - x_{102}^2 - x_{103}^2 - x_{107}^2 \\
&\quad - x_{108}^2 - x_{111}^2 - x_{114}^2 - x_{117}^2 - x_{120}^2 \\
&\quad - x_{121}^2 - x_{124}^2 - x_{127}^2 - x_{128}^2 - x_{132}^2 \\
&\quad - x_{135}^2 - x_{138}^2 + x_{141} - x_{211}^2 - x_{212}^2 \\
&\quad - x_{236}^2 \\
R_{156} &= - x_3 - x_3x_7 - x_3x_8 - x_3x_9 - x_3x_{10} \\
&\quad - x_3x_{11} - x_3x_{12} - x_3x_{13} - x_3x_{14} - x_3x_{15} \\
&\quad + x_{211} \\
R_{157} &= - x_{142}x_{214} - x_{214} + x_{223} \\
R_{158} &= - 1 + x_{213}^2 + x_{214}^2 - x_{223}^2 \\
R_{159} &= - x_{216} - x_{216}x_{217} + x_{224} \\
R_{160} &= - 1 + x_{215}^2 + x_{216}^2 - x_{224}^2 \\
R_{161} &= - x_{208}^2 + x_{225} \\
R_{162} &= x_{214} - 2x_{219}x_{225} - x_{225} \\
R_{163} &= - 4x_{214} - 8x_{214}x_{218} + x_{217} \\
R_{164} &= - x_{142} - x_{213}x_{220} + x_{217} \\
R_{165} &= - 1 - x_{207} - x_{213}x_{221} + x_{215} \\
R_{166} &= - x_{100} - 4x_{208}x_{222} + x_{216} \\
R_{167} &= - 3 + x_0^2 - 2x_0x_{196} + x_1^2 - 2x_1x_{184} \\
&\quad + x_2^2 + 2x_2x_{147} - 2x_2x_{156} + x_7^2 - 2x_7x_{162} \\
&\quad + x_8^2 - 2x_8x_{168} + x_9^2 - 2x_9x_{172} + x_{10}^2 \\
&\quad - 2x_{10}x_{176} + x_{11}^2 - 2x_{11}x_{180} + x_{12}^2 - 2x_{12}x_{188} \\
&\quad + x_{13}^2 - 2x_{13}x_{192} + x_{14}^2 - 2x_{14}x_{200} + x_{15}^2 \\
&\quad - 2x_{15}x_{204} - 2x_{100}x_{208} + 4x_{142} + 16x_{142}x_{212} - x_{142}x_{226} \\
&\quad + x_{146}^2 + x_{147}^2 - 2x_{147}x_{156} - 16x_{147}x_{212} - x_{149} \\
&\quad + x_{155}^2 + x_{161}^2 + 2x_{162} + x_{167}^2 + x_{171}^2 \\
&\quad + x_{175}^2 + x_{179}^2 + x_{183}^2 + x_{187}^2 + x_{191}^2 \\
&\quad + x_{195}^2 + x_{199}^2 + x_{203}^2 + x_{207}^2 + 2x_{208}^2 \\
&\quad - 16x_{212} - x_{230}^2 - x_{231}^2 - x_{232} - x_{232}^2 \\
R_{168} &= - x_{103}x_{227} + x_{128} \\
R_{169} &= - x_{108}x_{228} + x_{128} \\
R_{170} &= - x_{121}x_{229} + x_{128} \\
R_{171} &= - 1 + x_1 - x_2 - 2x_2x_{237} \\
R_{172} &= - 2 - x_{100}^2 - x_{147}^2 - x_{156}^2 + 2x_{162}
\end{aligned}
$$

$$
\begin{aligned}
&- x_{162}^2 - x_{168}^2 - x_{172}^2 - x_{176}^2 - x_{180}^2 \\
&- x_{184}^2 - x_{188}^2 - x_{192}^2 - x_{196}^2 - x_{200}^2 \\
&- x_{204}^2 + x_{208}^2 - x_{230}^2 - x_{231}^2 - x_{232} \\
&- x_{232}^2 \\
R_{173} = \; &- 1 - 4x_{105}^2 - 4x_{110}^2 - 4x_{113}^2 - 4x_{116}^2 \\
&- 4x_{119}^2 - 4x_{123}^2 - 4x_{126}^2 + x_{128}^2 - 4x_{130}^2 \\
&- 4x_{134}^2 - 4x_{137}^2 - 4x_{140}^2 - x_{233}^2 - x_{234}^2 \\
&- x_{235} - x_{235}^2
\end{aligned}
$$

B The complete 237-variable residual list

For the endpoint named `minimum_dimension`, define

$$P_{237}(x_0,\dots,x_{236}) = \sum_{j=0}^{172} S_j(x_0,\dots,x_{236})^2.$$

The coordinates below belong to this endpoint's own integral coordinate system. They are independent of the coordinates in the preceding appendix. All 173 residuals are printed in full.

$$\begin{aligned}
S_0 =& -x_1 - 8x_1x_{211} + x_6 + 8x_{211}\\
S_1 =& -x_1x_4 - x_2 + x_5x_6\\
S_2 =& -1 + x_0 - x_5x_6\\
S_3 =& 1 + x_2 - x_3x_{17} + x_{15}\\
S_4 =& -1 - x_7^2 - 2x_7x_8 - x_8^2 + x_{18}\\
S_5 =& -10x_7 + x_7^2 - 6x_7x_8 - 8x_7x_{18} - 20x_8\\
&- 11x_8^2 - 16x_8x_{18} - 15x_{18} - 4x_{18}^2 + x_{19}\\
S_6 =& -2 + 2x_7 - x_7^2 - 4x_7x_{19} - 4x_{19}\\
&- 4x_{19}^2 + x_{20}\\
S_7 =& 1 + 6x_7 - 36x_7^2 - 24x_7x_{20} + 2x_{20}\\
&- 4x_{20}^2 + x_{21}\\
S_8 =& -2 - x_8 - x_8^2 - 2x_8x_9 - x_9\\
&- x_9^2 + x_{22}\\
S_9 =& -8 + 7x_7 - x_7^2 + 4x_7x_9 - 4x_7x_{22}\\
&- 4x_9 + 4x_{22} + x_{23} - x_{45}\\
S_{10} =& -1 + x_7 - x_7^2 - 2x_7x_9 + x_9\\
&- x_9^2 + x_{24}\\
S_{11} =& 23 + 16x_7 - 8x_7x_8 - 8x_7x_9 - 16x_7x_{18}\\
&- 8x_7x_{24} + 19x_8 + 9x_8^2 - 36x_8x_9 - 12x_8x_{24}\\
&+ 16x_9 + 8x_9^2 - 24x_9x_{18} - 8x_{18}x_{24} - x_{19}\\
&+ x_{25} - x_{30}\\
S_{12} =& -4x_{23}^2 - 8x_{23}x_{25} - 4x_{25}^2 + x_{26}\\
S_{13} =& -2 - 8x_7 + 8x_7x_{22} + 8x_7x_{24} - 8x_8\\
&+ 8x_8x_{22} + 8x_8x_{24} - 4x_{18} + 8x_{22} - 4x_{22}^2\\
&- 8x_{22}x_{24} + 8x_{24} - 4x_{24}^2 + x_{27}\\
S_{14} =& 6 + 4x_7 + 32x_7x_8 + 16x_7x_{18} + 16x_7x_{24}\\
&+ 8x_7x_{27} + 24x_8 - 32x_8x_{24} - 16x_8x_{27} + 12x_{18}\\
&- 16x_{18}x_{24} - 8x_{18}x_{27} - x_{19} + 8x_{24} - 16x_{24}^2\\
&- 16x_{24}x_{27} + 4x_{27} - 4x_{27}^2 + x_{28}\\
S_{15} =& 22 - 5x_7 - x_7^2 - 8x_7x_8 - 12x_7x_9\\
&- 4x_7x_{22} - 8x_8 - 16x_8x_9 - 16x_8x_{22} - 20x_9\\
&- 20x_9^2 - 24x_9x_{22} - 28x_{22} - 4x_{22}^2 + x_{29}\\
S_{16} =& 19 - 25x_7 - 7x_7^2 - 8x_7x_8 - 30x_7x_9\\
&- 16x_7x_{24} + 3x_8 + 5x_8^2 - 4x_8x_{24} - 27x_9\\
&- 21x_9^2 - 24x_9x_{24} - 6x_{22} - 17x_{24} - 4x_{24}^2\\
&+ x_{30}\\
S_{17} =& 2 - 4x_{29}^2 - 8x_{29}x_{30} - 4x_{30}^2 + x_{31}\\
S_{18} =& -19x_7 + 8x_7^2 + 8x_7x_{18} + 8x_7x_{22} - 46x_8\\
&- 24x_8^2 + 24x_8x_{18} + 24x_8x_{22} - 2x_9 + 4x_{18}
\end{aligned}$$

$$
\begin{aligned}
&\quad - 4x_{18}^2 - 8x_{18}x_{22} + 14x_{22} - 4x_{22}^2 - x_{24} \\
&\quad + x_{32} \\
S_{19} = &\ -3 - 6x_7 - 4x_9 - 2x_{24} - 4x_{32} \\
&\quad - 4x_{32}^2 + x_{33} \\
S_{20} = &\ 8 + 42x_7 + 7x_7^2 + 28x_7x_{19} + 48x_8 \\
&\quad - 32x_8^2 - 32x_8x_{18} + 64x_8x_{19} - 40x_{18}^2 + 32x_{18}x_{19} \\
&\quad + 14x_{19} - 4x_{19}^2 + x_{34} + 2x_{37} \\
S_{21} = &\ 1 + 6x_7 - 36x_7^2 - 24x_7x_{34} + 2x_{34} \\
&\quad - 4x_{34}^2 + x_{35} \\
S_{22} = &\ -x_8^2 - 4x_8x_{35} - 4x_{35}^2 + x_{36} \\
S_{23} = &\ -6 - 2x_7 - 4x_8 - 14x_{18} - 16x_{18}^2 \\
&\quad + x_{37} \\
S_{24} = &\ 24 + 21x_8 - 21x_8^2 - 18x_8x_9 - 24x_8x_{22} \\
&\quad - 24x_8x_{37} + 3x_9 - 9x_9^2 - 16x_9x_{37} + 3x_{22} \\
&\quad - 12x_{22}^2 - 8x_{22}x_{37} + 4x_{37} - 4x_{37}^2 + x_{38} \\
&\quad + 2x_{45} \\
S_{25} = &\ 16 - 10x_7 - 20x_7^2 - 16x_7x_9 - 24x_7x_{24} \\
&\quad - 24x_7x_{38} - 12x_9 - 16x_9x_{24} - 16x_9x_{38} - 14x_{24} \\
&\quad - 4x_{24}^2 - 8x_{24}x_{38} + 4x_{38} - 4x_{38}^2 + x_{39} \\
S_{26} = &\ -x_9^2 - 2x_9x_{10} - x_{10}^2 + x_{40} \\
S_{27} = &\ 2 + x_7^2 - 4x_7x_{40} + 2x_9 - 2x_9^2 \\
&\quad - 8x_9x_{40} - 2x_{24} + 4x_{40} - 4x_{40}^2 + x_{41} \\
S_{28} = &\ -31 - 16x_9 - 16x_9x_{22} + 16x_9x_{24} - 8x_{22} \\
&\quad + 12x_{22}^2 - 8x_{22}x_{24} + 24x_{24} - 4x_{24}^2 + x_{42} \\
&\quad - 4x_{45} \\
S_{29} = &\ -4x_9 - 16x_{24}^2 - 32x_{24}x_{40} - 16x_{24}x_{41} - 16x_{24}x_{42} \\
&\quad - 4x_{40} - 16x_{40}^2 - 16x_{40}x_{41} - 16x_{40}x_{42} - 2x_{41} \\
&\quad - 4x_{41}^2 - 8x_{41}x_{42} - 4x_{42}^2 + x_{43} \\
S_{30} = &\ -16x_7^2 + 8x_7x_9 + 32x_7x_{24} + 16x_7x_{27} + x_9 \\
&\quad - x_9^2 - 8x_9x_{24} - 4x_9x_{27} - 16x_{24}^2 - 16x_{24}x_{27} \\
&\quad - 4x_{27}^2 + x_{44} \\
S_{31} = &\ -1 - 8x_9 - 4x_9^2 + 8x_9x_{22} + 8x_{22} \\
&\quad - 4x_{22}^2 + x_{45} \\
S_{32} = &\ -1 - 9x_9^2 + 24x_9x_{22} + 12x_9x_{45} - 16x_{22}^2 \\
&\quad - 16x_{22}x_{45} - 4x_{45}^2 + x_{46} \\
S_{33} = &\ 1 + x_8 - x_8^2 - 4x_8x_9 - 4x_8x_{46} \\
&\quad - 4x_9^2 - 8x_9x_{46} - 4x_{46}^2 + x_{47} \\
S_{34} = &\ -1 - 4x_{47} - 4x_{47}^2 + x_{48} \\
S_{35} = &\ -1 - x_9^2 - 8x_9x_{47} - 4x_9x_{48} - 16x_{47}^2 \\
&\quad - 16x_{47}x_{48} - 4x_{48}^2 + x_{49} \\
S_{36} = &\ 2 + x_8 - x_8^2 - 4x_8x_9 - 4x_8x_{49} \\
&\quad - 4x_9^2 - 8x_9x_{49} - 4x_{49}^2 + x_{50} \\
S_{37} = &\ -4x_{44}^2 - 8x_{44}x_{50} - 4x_{50}^2 + x_{51} \\
S_{38} = &\ -9 + 7x_7 - x_7^2 - 8x_7x_{44} - 4x_7x_{51} \\
&\quad + 24x_{44} - 16x_{44}^2 - 16x_{44}x_{51} + 12x_{51} - 4x_{51}^2 \\
&\quad + x_{52} \\
S_{39} = &\ -1 - 4x_{52} - 4x_{52}^2 + x_{53}
\end{aligned}
$$

$$
\begin{aligned}
S_{40} = & -x_9^2 - 8x_9x_{52} - 4x_9x_{53} - 16x_{52}^2 - 16x_{52}x_{53} \\
& - 4x_{53}^2 + x_{54} \\
S_{41} = & -x_7 - x_7^2 - 4x_7x_9 - 4x_7x_{54} - 4x_9 \\
& - 4x_9^2 - 8x_9x_{54} - 4x_{54} - 4x_{54}^2 + x_{55} \\
S_{42} = & -10 - 12x_{55} - 4x_{55}^2 + x_{56} \\
S_{43} = & \, 2 + 4x_9 + 16x_9x_{55} + 8x_9x_{56} - 4x_{22} \\
& - 16x_{22}x_{55} - 8x_{22}x_{56} - x_{45} + 8x_{55} - 16x_{55}^2 \\
& - 16x_{55}x_{56} + 4x_{56} - 4x_{56}^2 + x_{57} \\
S_{44} = & \, x_8 - x_8^2 + 8x_8x_9 - 8x_8x_{22} - 4x_8x_{57} \\
& - 16x_9^2 + 32x_9x_{22} + 16x_9x_{57} - 16x_{22}^2 - 16x_{22}x_{57} \\
& - 4x_{57}^2 + x_{58} \\
S_{45} = & -10 - 12x_{50} - 4x_{50}^2 + x_{59} \\
S_{46} = & \, 1 - 16x_{50}^2 - 16x_{50}x_{58} - 16x_{50}x_{59} - 4x_{58}^2 \\
& - 8x_{58}x_{59} - 4x_{59}^2 + x_{60} \\
S_{47} = & \, x_9 - x_9^2 - 8x_9x_{58} - 4x_9x_{60} - 16x_{58}^2 \\
& - 16x_{58}x_{60} - 4x_{60}^2 + x_{61} \\
S_{48} = & -6x_7 - 3x_7^2 - 18x_7x_8 - 16x_8 - 27x_8^2 \\
& + 4x_8x_9 + 3x_9 + 3x_9^2 - 4x_9x_{18} - 19x_{18} \\
& - 8x_{18}^2 + x_{19} + x_{62} - 4x_{75} \\
S_{49} = & \, 1 - 8x_7 - 4x_7x_9 + 8x_7x_{62} - 16x_8 \\
& - 8x_8x_9 + 16x_8x_{62} + x_9 + x_9^2 + 4x_9x_{18} \\
& + 8x_{18} - 8x_{18}x_{62} + 3x_{62} - 4x_{62}^2 + x_{63} \\
S_{50} = & -2 - 12x_7 - 16x_7x_9 + 16x_7x_{61} + 16x_7x_{63} \\
& - 24x_8 - 32x_8x_9 + 32x_8x_{61} + 32x_8x_{63} + 4x_9 \\
& + 4x_9^2 + 16x_9x_{18} + 12x_{18} - 16x_{18}x_{61} - 16x_{18}x_{63} \\
& - 4x_{61}^2 - 8x_{61}x_{63} - 4x_{62} - 2x_{63} - 4x_{63}^2 \\
& + x_{64} \\
S_{51} = & -2 - 4x_{64} - 4x_{64}^2 + x_{65} \\
S_{52} = & -x_2x_{12} + x_{66} \\
S_{53} = & \, x_{11} - x_{12} - 2x_{21}x_{66} - x_{67} \\
S_{54} = & \, x_{11} - x_{12} - 4x_{23}x_{66} - 2x_{26}x_{66} - x_{68} \\
S_{55} = & -2x_{28}x_{66} + 2x_{43}x_{66} - 2x_{66} - x_{69} + x_{74} \\
S_{56} = & \, x_{11} - x_{12} - 4x_{29}x_{66} - 2x_{31}x_{66} - x_{70} \\
S_{57} = & \, x_{11} - x_{12} - 2x_{33}x_{66} + 6x_{66} - x_{71} \\
S_{58} = & \, x_{11} - x_{12} - 4x_{35}x_{66} - 2x_{36}x_{66} - x_{72} \\
S_{59} = & \, x_{11} - x_{12} - 2x_{39}x_{66} - x_{73} \\
S_{60} = & \, x_{11} - x_{12} - 2x_{43}x_{66} - 4x_{66} - x_{74} \\
S_{61} = & -x_7x_8 + x_7x_9 + x_8 + x_8x_9 - x_9 \\
& - x_9^2 + x_{75} \\
S_{62} = & \, 1 + 2x_{10} + x_{10}^2 - x_{75}^2 + x_{76} \\
S_{63} = & -x_{76}^2 + x_{77} \\
S_{64} = & -4x_{64}x_{66} - 2x_{65}x_{66} + 2x_{66} + x_{78} \\
S_{65} = & -x_{11}^2 + 2x_{11}x_{12} + 2x_{11}x_{78} - x_{12}^2 - 2x_{12}x_{78} \\
& - x_{78}^2 + x_{79} \\
S_{66} = & -1 - x_2x_8 + x_{80} \\
S_{67} = & -x_{12}x_{80} + x_{82}x_{83} + x_{84} \\
S_{68} = & -1 + x_2 - x_2x_7 + x_{81} \\
S_{69} = & -1 + 2x_2 + 2x_2x_7 + 4x_2x_8 - 2x_2x_{18}
\end{aligned}
$$

$$
\begin{aligned}
&+ x_{82}\\
S_{70} =\ & - x_{11}x_{81} + x_{83}\\
S_{71} =\ & - 6x_{2}x_{12} + x_{11}x_{80} + 12x_{12} - 8x_{12}x_{80} - 4x_{12}x_{81}\\
&- x_{12}x_{82} + x_{85}\\
S_{72} =\ & - x_{11}x_{67} + x_{12}x_{67} + x_{86}\\
S_{73} =\ & - x_{68}x_{86} + x_{87}\\
S_{74} =\ & - x_{69}x_{87} + x_{88}\\
S_{75} =\ & - x_{70}x_{88} + x_{89}\\
S_{76} =\ & - x_{71}x_{89} + x_{90}\\
S_{77} =\ & - x_{72}x_{90} + x_{91}\\
S_{78} =\ & - x_{73}x_{91} + x_{92}\\
S_{79} =\ & - x_{74}x_{92} + x_{93}\\
S_{80} =\ & - x_{84}x_{93} + x_{94}\\
S_{81} =\ & - x_{85}x_{94} + x_{95}\\
S_{82} =\ & - x_{95}^{2} + x_{96}\\
S_{83} =\ & - x_{1} - x_{1}x_{15} + x_{15} + x_{97}\\
S_{84} =\ & - x_{13}x_{97} + x_{98}\\
S_{85} =\ & - 2x_{0}^{2} + 2x_{0}x_{11} + 2x_{0}x_{12} + 2x_{0}x_{98} + 2x_{0}x_{99}\\
&- x_{11}^{2} - 2x_{11}x_{98} - x_{12}^{2} - 2x_{12}x_{99} + x_{16}x_{100}\\
&- x_{77}x_{96} - x_{79}x_{96} - x_{98}^{2} - x_{99}^{2}\\
S_{86} =\ & - x_{1}x_{14} + x_{14} - x_{14}x_{97} + x_{99}\\
S_{87} =\ & - x_{103} - 4x_{103}x_{235} + x_{105}\\
S_{88} =\ & - x_{3}x_{127} + x_{101}x_{226} - x_{104} - x_{105}x_{127}\\
S_{89} =\ & x_{106}x_{227} - x_{108}x_{130} - x_{109}\\
S_{90} =\ & x_{110}x_{227} - x_{111}x_{130} - x_{112}\\
S_{91} =\ & x_{113}x_{227} - x_{114}x_{130} - x_{115}\\
S_{92} =\ & x_{116}x_{227} - x_{117}x_{130} - x_{118}\\
S_{93} =\ & x_{119}x_{228} - x_{121}x_{130} - x_{122}\\
S_{94} =\ & x_{123}x_{228} - x_{124}x_{130} - x_{125}\\
S_{95} =\ & - x_{16}x_{127} + x_{130}\\
S_{96} =\ & x_{126} - x_{128}x_{130} - x_{129}\\
S_{97} =\ & - x_{130}x_{132} + x_{131} - x_{133}\\
S_{98} =\ & - x_{130}x_{135} + x_{134}x_{226} - x_{136}\\
S_{99} =\ & - x_{130}x_{138} + x_{137}x_{227} - x_{139}\\
S_{100} =\ & x_{140} - 2x_{140}x_{142} + x_{143}\\
S_{101} =\ & - x_{140}x_{143} - x_{143} + x_{144}\\
S_{102} =\ & x_{143}^{2} - x_{144}^{2} + x_{164}\\
S_{103} =\ & - x_{141}x_{146} - x_{146} + x_{147}\\
S_{104} =\ & 2x_{145} + x_{145}^{2} + x_{146}^{2} - x_{147}^{2}\\
S_{105} =\ & 1 - 4x_{141} - 16x_{141}x_{211} + x_{148} + 16x_{211}\\
&+ 64x_{211}^{2}\\
S_{106} =\ & - 1 + x_{2} + x_{141}x_{146} - x_{145} - x_{146}\\
&- 8x_{146}x_{211} + x_{148}x_{149}\\
S_{107} =\ & 1 - 2x_{3} + x_{3}^{2} - 2x_{3}x_{141} + x_{150}\\
S_{108} =\ & - x_{3}x_{195} + x_{16} + x_{150}x_{151} - x_{194} + x_{196}\\
S_{109} =\ & - 1 + 2x_{141}x_{210} - x_{152} + 2x_{210} - x_{210}^{2}\\
S_{110} =\ & 1 + x_{152}x_{153} - x_{194} - x_{195}x_{210} + x_{196}\\
&+ 4x_{235}\\
S_{111} =\ & - x_{141}x_{155} - x_{155} + x_{156}\\
S_{112} =\ & 2x_{154} + x_{154}^{2} + x_{155}^{2} - x_{156}^{2}
\end{aligned}
$$

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\begin{aligned}
S_{113} &= 1 - 4x_{141} - 8x_{141}x_{235} + x_{157} + 8x_{235} \\
&\quad + 16x_{235}^2 \\
S_{114} &= -1 + x_{101} - x_{154} - 2x_{155} - 4x_{155}x_{235} \\
&\quad + x_{156} + x_{157}x_{158} \\
S_{115} &= -1 + x_{102} - 2x_{141}x_{159} - x_{145} - x_{146} \\
&\quad - 4x_{146}x_{235} + x_{147} + x_{157}x_{159} + x_{159} + 8x_{159}x_{235} \\
S_{116} &= 1 + x_{141} - x_{141}x_{161} - x_{161} + x_{162} \\
S_{117} &= 1 + 2x_{160} + x_{160}^2 - 2x_{161} + x_{161}^2 \\
&\quad - x_{162}^2 \\
S_{118} &= 1 + x_{106} + x_{157}x_{163} - x_{160} - 2x_{161} \\
&\quad - 4x_{161}x_{235} + x_{162} + 4x_{235} \\
S_{119} &= -2x_{141} - x_{141}^2 + x_{164} \\
S_{120} &= -1 + x_{107} + x_{141}x_{147} - 2x_{141}x_{165} - x_{145} \\
&\quad - 4x_{145}x_{235} - x_{146}x_{164} - 4x_{147}x_{235} + x_{157}x_{165} + x_{165} \\
&\quad + 8x_{165}x_{235} - 4x_{235} \\
S_{121} &= -x_{141}x_{167} - x_{167} + x_{168} \\
S_{122} &= 2x_{166} + x_{166}^2 + x_{167}^2 - x_{168}^2 \\
S_{123} &= -1 + x_{110} + x_{157}x_{169} - x_{166} - 2x_{167} \\
&\quad - 4x_{167}x_{235} + x_{168} \\
S_{124} &= -x_{141}x_{171} - x_{171} + x_{172} \\
S_{125} &= 2x_{170} + x_{170}^2 + x_{171}^2 - x_{172}^2 \\
S_{126} &= -1 + x_{113} + x_{157}x_{173} - x_{170} - 2x_{171} \\
&\quad - 4x_{171}x_{235} + x_{172} \\
S_{127} &= -x_{141}x_{175} - x_{175} + x_{176} \\
S_{128} &= 2x_{174} + x_{174}^2 + x_{175}^2 - x_{176}^2 \\
S_{129} &= -1 + x_{116} + x_{157}x_{177} - x_{174} - 2x_{175} \\
&\quad - 4x_{175}x_{235} + x_{176} \\
S_{130} &= -x_{141}x_{179} - x_{179} + x_{180} \\
S_{131} &= 2x_{178} + x_{178}^2 + x_{179}^2 - x_{180}^2 \\
S_{132} &= -1 + x_{119} + x_{157}x_{181} - x_{178} - 2x_{179} \\
&\quad - 4x_{179}x_{235} + x_{180} \\
S_{133} &= -x_{141}x_{183} - x_{183} + x_{184} \\
S_{134} &= 2x_{182} + x_{182}^2 + x_{183}^2 - x_{184}^2 \\
S_{135} &= -1 + x_{120} - 2x_{141}x_{185} + x_{157}x_{185} - x_{182} \\
&\quad - x_{183} - 4x_{183}x_{235} + x_{184} + x_{185} + 8x_{185}x_{235} \\
S_{136} &= -x_{141}x_{187} - x_{187} + x_{188} \\
S_{137} &= 2x_{186} + x_{186}^2 + x_{187}^2 - x_{188}^2 \\
S_{138} &= -1 + x_{123} + x_{157}x_{189} - x_{186} - 2x_{187} \\
&\quad - 4x_{187}x_{235} + x_{188} \\
S_{139} &= -x_{141}x_{191} - x_{191} + x_{192} \\
S_{140} &= 2x_{190} + x_{190}^2 + x_{191}^2 - x_{192}^2 \\
S_{141} &= -1 + x_{126} + x_{157}x_{193} - x_{190} - 2x_{191} \\
&\quad - 4x_{191}x_{235} + x_{192} \\
S_{142} &= 1 - x_{141}x_{195} - x_{195} + x_{196} \\
S_{143} &= -1 + 2x_{194} + x_{194}^2 + x_{195}^2 - 2x_{196} \\
&\quad - x_{196}^2 \\
S_{144} &= x_{127} - 2x_{141}x_{197} + x_{157}x_{197} - x_{194} - x_{195} \\
&\quad - 4x_{195}x_{235} + x_{196} + x_{197} + 8x_{197}x_{235} \\
S_{145} &= -x_{141}x_{199} - x_{199} + x_{200} \\
S_{146} &= 2x_{198} + x_{198}^2 + x_{199}^2 - x_{200}^2
\end{aligned}
$$

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\begin{aligned}
S_{147} = & -1 + x_{131} + x_{157}x_{201} - x_{198} - 2x_{199} \\
& - 4x_{199}x_{235} + x_{200} \\
S_{148} = & -x_{141}x_{203} - x_{203} + x_{204} \\
S_{149} = & 2x_{202} + x_{202}^2 + x_{203}^2 - x_{204}^2 \\
S_{150} = & -1 + x_{134} + x_{157}x_{205} - x_{202} - 2x_{203} \\
& - 4x_{203}x_{235} + x_{204} \\
S_{151} = & -x_{141}x_{207} - x_{207} + x_{208} \\
S_{152} = & 2x_{206} + x_{206}^2 + x_{207}^2 - x_{208}^2 \\
S_{153} = & -1 + x_{137} + x_{157}x_{209} - x_{206} - 2x_{207} \\
& - 4x_{207}x_{235} + x_{208} \\
S_{154} = & -30 - x_0^2 + 2x_1 - x_1^2 - x_2^2 \\
& - x_3^2 + 6x_7 + 2x_7x_8 + 4x_8 + 2x_9 \\
& - x_9^2 + 2x_{10} - x_{10}^2 + 2x_{11} - x_{11}^2 \\
& + 2x_{12} - x_{12}^2 + 2x_{13} - x_{13}^2 + 2x_{14} \\
& - x_{14}^2 + 2x_{15} - x_{15}^2 - x_{16}^2 + 7x_{18} \\
& + 4x_{21} + 8x_{23} + 4x_{26} + 4x_{28} + 8x_{29} \\
& + 4x_{31} + 4x_{33} + 8x_{35} + 4x_{36} + 4x_{39} \\
& + 4x_{43} + 8x_{64} + 4x_{65} - x_{101}^2 - x_{102}^2 \\
& - x_{106}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 - x_{116}^2 \\
& - x_{119}^2 - x_{120}^2 - x_{123}^2 - x_{126}^2 - x_{127}^2 \\
& - x_{131}^2 - x_{134}^2 - x_{137}^2 + x_{140} + x_{146}^2 \\
& + x_{155}^2 - 2x_{161} + x_{161}^2 + x_{167}^2 + x_{171}^2 \\
& + x_{175}^2 + x_{179}^2 + x_{183}^2 + x_{187}^2 + x_{191}^2 \\
& + x_{195}^2 + x_{199}^2 + x_{203}^2 - x_{207}^2 - x_{210}^2 \\
& - x_{211}^2 + x_{229}^2 + x_{230}^2 + x_{231} + x_{231}^2 \\
& - x_{235}^2 \\
S_{155} = & -x_3 - x_3x_7 - x_3x_8 - x_3x_9 - x_3x_{10} \\
& - x_3x_{11} - x_3x_{12} - x_3x_{13} - x_3x_{14} - x_3x_{15} \\
& + x_{210} \\
S_{156} = & -x_{141}x_{213} - x_{213} + x_{222} \\
S_{157} = & -1 + x_{212}^2 + x_{213}^2 - x_{222}^2 \\
S_{158} = & -x_{215} - x_{215}x_{216} + x_{223} \\
S_{159} = & -1 + x_{214}^2 + x_{215}^2 - x_{223}^2 \\
S_{160} = & -x_{207}^2 + x_{224} \\
S_{161} = & x_{213} - 2x_{218}x_{224} - x_{224} \\
S_{162} = & -4x_{213} - 8x_{213}x_{217} + x_{216} \\
S_{163} = & -x_{141} - x_{212}x_{219} + x_{216} \\
S_{164} = & -1 - x_{206} - x_{212}x_{220} + x_{214} \\
S_{165} = & -x_7 - x_8 - 4x_{18} - 2x_{21} - 4x_{23} \\
& - 2x_{26} - 2x_{28} - 4x_{29} - 2x_{31} - 2x_{33} \\
& - 4x_{35} - 2x_{36} - 2x_{39} - 2x_{43} - 4x_{64} \\
& - 2x_{65} - 4x_{207}x_{221} + x_{215} \\
S_{166} = & -3 + x_0^2 - 2x_0x_{195} + x_1^2 - 2x_1x_{183} \\
& + x_2^2 + 2x_2x_{146} - 2x_2x_{155} + x_7^2 - 2x_7x_{161} \\
& - 2x_7x_{207} + x_8^2 - 2x_8x_{167} - 2x_8x_{207} + x_9^2 \\
& - 2x_9x_{171} + x_{10}^2 - 2x_{10}x_{175} + x_{11}^2 - 2x_{11}x_{179} \\
& + x_{12}^2 - 2x_{12}x_{187} + x_{13}^2 - 2x_{13}x_{191} + x_{14}^2 \\
& - 2x_{14}x_{199} + x_{15}^2 - 2x_{15}x_{203} - 8x_{18}x_{207} - 4x_{21}x_{207} \\
& - 8x_{23}x_{207} - 4x_{26}x_{207} - 4x_{28}x_{207} - 8x_{29}x_{207} - 4x_{31}x_{207}
\end{aligned}
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\begin{aligned}
&- 4x_{33}x_{207} - 8x_{35}x_{207} - 4x_{36}x_{207} - 4x_{39}x_{207} - 4x_{43}x_{207} \\
&- 8x_{64}x_{207} - 4x_{65}x_{207} + 4x_{141} + 16x_{141}x_{211} - x_{141}x_{225} \\
&+ x_{145}^2 + x_{146}^2 - 2x_{146}x_{155} - 16x_{146}x_{211} - x_{148} \\
&+ x_{154}^2 + x_{160}^2 + 2x_{161} + x_{166}^2 + x_{170}^2 \\
&+ x_{174}^2 + x_{178}^2 + x_{182}^2 + x_{186}^2 + x_{190}^2 \\
&+ x_{194}^2 + x_{198}^2 + x_{202}^2 + x_{206}^2 + 2x_{207}^2 \\
&- 16x_{211} - x_{229}^2 - x_{230}^2 - x_{231} - x_{231}^2 \\
S_{167} = \; &- x_{102}x_{226} + x_{127} \\
S_{168} = \; &- x_{107}x_{227} + x_{127} \\
S_{169} = \; &- x_{120}x_{228} + x_{127} \\
S_{170} = \; &- 1 + x_1 - x_2 - 2x_2x_{236} \\
S_{171} = \; &- 1 - 8x_7x_{18} - 4x_7x_{21} - 8x_7x_{23} - 4x_7x_{26} \\
&- 4x_7x_{28} - 8x_7x_{29} - 4x_7x_{31} - 4x_7x_{33} - 8x_7x_{35} \\
&- 4x_7x_{36} - 4x_7x_{39} - 4x_7x_{43} - 8x_7x_{64} - 4x_7x_{65} \\
&- 8x_8x_{18} - 4x_8x_{21} - 8x_8x_{23} - 4x_8x_{26} - 4x_8x_{28} \\
&- 8x_8x_{29} - 4x_8x_{31} - 4x_8x_{33} - 8x_8x_{35} - 4x_8x_{36} \\
&- 4x_8x_{39} - 4x_8x_{43} - 8x_8x_{64} - 4x_8x_{65} - x_{18} \\
&- 16x_{18}^2 - 16x_{18}x_{21} - 32x_{18}x_{23} - 16x_{18}x_{26} - 16x_{18}x_{28} \\
&- 32x_{18}x_{29} - 16x_{18}x_{31} - 16x_{18}x_{33} - 32x_{18}x_{35} - 16x_{18}x_{36} \\
&- 16x_{18}x_{39} - 16x_{18}x_{43} - 32x_{18}x_{64} - 16x_{18}x_{65} - 4x_{21}^2 \\
&- 16x_{21}x_{23} - 8x_{21}x_{26} - 8x_{21}x_{28} - 16x_{21}x_{29} - 8x_{21}x_{31} \\
&- 8x_{21}x_{33} - 16x_{21}x_{35} - 8x_{21}x_{36} - 8x_{21}x_{39} - 8x_{21}x_{43} \\
&- 16x_{21}x_{64} - 8x_{21}x_{65} - 16x_{23}^2 - 16x_{23}x_{26} - 16x_{23}x_{28} \\
&- 32x_{23}x_{29} - 16x_{23}x_{31} - 16x_{23}x_{33} - 32x_{23}x_{35} - 16x_{23}x_{36} \\
&- 16x_{23}x_{39} - 16x_{23}x_{43} - 32x_{23}x_{64} - 16x_{23}x_{65} - 4x_{26}^2 \\
&- 8x_{26}x_{28} - 16x_{26}x_{29} - 8x_{26}x_{31} - 8x_{26}x_{33} - 16x_{26}x_{35} \\
&- 8x_{26}x_{36} - 8x_{26}x_{39} - 8x_{26}x_{43} - 16x_{26}x_{64} - 8x_{26}x_{65} \\
&- 4x_{28}^2 - 16x_{28}x_{29} - 8x_{28}x_{31} - 8x_{28}x_{33} - 16x_{28}x_{35} \\
&- 8x_{28}x_{36} - 8x_{28}x_{39} - 8x_{28}x_{43} - 16x_{28}x_{64} - 8x_{28}x_{65} \\
&- 16x_{29}^2 - 16x_{29}x_{31} - 16x_{29}x_{33} - 32x_{29}x_{35} - 16x_{29}x_{36} \\
&- 16x_{29}x_{39} - 16x_{29}x_{43} - 32x_{29}x_{64} - 16x_{29}x_{65} - 4x_{31}^2 \\
&- 8x_{31}x_{33} - 16x_{31}x_{35} - 8x_{31}x_{36} - 8x_{31}x_{39} - 8x_{31}x_{43} \\
&- 16x_{31}x_{64} - 8x_{31}x_{65} - 4x_{33}^2 - 16x_{33}x_{35} - 8x_{33}x_{36} \\
&- 8x_{33}x_{39} - 8x_{33}x_{43} - 16x_{33}x_{64} - 8x_{33}x_{65} - 16x_{35}^2 \\
&- 16x_{35}x_{36} - 16x_{35}x_{39} - 16x_{35}x_{43} - 32x_{35}x_{64} - 16x_{35}x_{65} \\
&- 4x_{36}^2 - 8x_{36}x_{39} - 8x_{36}x_{43} - 16x_{36}x_{64} - 8x_{36}x_{65} \\
&- 4x_{39}^2 - 8x_{39}x_{43} - 16x_{39}x_{64} - 8x_{39}x_{65} - 4x_{43}^2 \\
&- 16x_{43}x_{64} - 8x_{43}x_{65} - 16x_{64}^2 - 16x_{64}x_{65} - 4x_{65}^2 \\
&- x_{146}^2 - x_{155}^2 + 2x_{161} - x_{161}^2 - x_{167}^2 \\
&- x_{171}^2 - x_{175}^2 - x_{179}^2 - x_{183}^2 - x_{187}^2 \\
&- x_{191}^2 - x_{195}^2 - x_{199}^2 - x_{203}^2 + x_{207}^2 \\
&- x_{229}^2 - x_{230}^2 - x_{231} - x_{231}^2 \\
S_{172} = \; &- 1 - 4x_{104}^2 - 4x_{109}^2 - 4x_{112}^2 - 4x_{115}^2 \\
&- 4x_{118}^2 - 4x_{122}^2 - 4x_{125}^2 + x_{127}^2 - 4x_{129}^2 \\
&- 4x_{133}^2 - 4x_{136}^2 - 4x_{139}^2 - x_{232}^2 - x_{233}^2 \\
&- x_{234} - x_{234}^2
\end{aligned}
$$